

**STUDENT INTERNSHIP PROGRAM(SIP)
REPORT
JUNE - JULY
(2023-24)**

Name of the Industry- TATA (Forage)

Submitted By

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PRN-202301040037

Roll no . 62 Batch . A2

SCHOOL OF COMPUTER ENGINEERING

ACADEMIC YEAR 2023-24

MIT Academy of Engineering Alandi, Pune 412105

MIT ACADEMY OF ENGINEERING ALANDI(D), PUNE

CERTIFICATE

This is to certify that the “**Summer Internship Program (SIP)**” has **been successfully in the course**. So I am Vaishnavi Nehare . PRN-202301040037 has finished the work and submitting report of SIP during this academic year Sep 2024- 25.

School-Internship Coordinator (SIP)

Ms. Prajakata Dashrath

Head of Department

Dr . Pramod D. Ganjewar

School Dean

Mrs. Rajeshwari M. Goudar

Internship Certificate :




Forage
Inspiring and empowering
future professionals

Vaishnavi Tara Somnath Nehare

Data Visualisation: Empowering Business with Effective Insights

Certificate of Completion
July 27th, 2024

Over the period of June 2024 to July 2024, Vaishnavi Tara Somnath Nehare has completed practical tasks in:

- Framing the Business Scenario
- Choosing the Right Visuals
- Creating Effective Visuals
- Communicating Insights and Analysis



Tom Brunskill
CEO, Co-Founder of
Forage

Enrolment Verification Code BzLrH3m9m9Yqkv477
User Verification Code NRguPzaykdmnBk95A
Issued by **Forage**

Additional Internship



Corporate
CISCO Social
Networkability
Certificate of Course Completion

Cisco Networking Academy

Introduction to Cybersecurity

For completing the Cisco Networking Academy® Introduction to Cybersecurity course, and demonstrating the ability to explain the following:

- Global implications of cyber threats
- Ways in which networks are vulnerable to attack
- Impact of cyber attacks on industries

- Cisco's approach to threat detection and defense
- Why cybersecurity is a growing profession
- Opportunities available for pursuing network security certifications

VAISHNAVI NEHARE
Student

MIT Academy of Engineering
Academy Name

India 28 Jun 2024

Location: _____ Date: _____



Anuradha Pawar
Instructor

Mentor Signature

Certificate :



Acknowledgment

I am expressing my sincere gratitude to everyone who supported and guided me throughout my internship. Firstly, I would like to thank **TATA Forage** for allowing me to gain practical experience and learn from experts in the field.

I am deeply thankful to my faculty mentor, Madhavi Nimkar, for their valuable guidance and encouragement during the internship period. Their insights and feedback were instrumental in shaping my learning experience.

I also extend my thanks to the School of Humanities and Engineering Sciences and the internship coordinator, Anuradha Pawar, for their continuous support and coordination throughout the internship program.

Lastly, I am grateful to my peers and colleagues for making this internship an enriching and collaborative experience.

Thank you to everyone who has contributed to my professional development during the internship.

Sincerely,

Vaishnavi Nehare.

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1. Introduction

1.1. About Forage

Forage is a virtual internship platform that offers self-paced, free, and flexible internship opportunities to students and young professionals. The platform provides a range of simulations and projects that mimic real-world work experiences, allowing users to gain valuable skills and build their portfolios. Forage also provides resources and guides to help users improve their skills and prepare for the workforce. Using Forage can help users gain practical work experience, build their portfolios, and develop valuable skills. The platform also provides an opportunity for users to learn from their mistakes and receive feedback on their work. Additionally, Forage can help users increase their chances of getting hired by top companies.

1.2. Internship Overview

My internship program on "Data Visualisation: Empowering Business with Effective Insights" was completed as a part of a structured learning course designed to enhance my data visualization skills. The program focused on developing the ability to transform raw data into meaningful insights that drive business decisions. The internship consisted of the following key tasks:

- **Framing the Business Scenario:** Focused on understanding the business context and identifying key data points to create effective visuals.
- **Choosing the Right Visuals:** Involved selecting the most suitable visual representations for different data types, ensuring clarity in communication.
- **Creating Effective Visuals:** Provided hands-on experience in using visualization tools to turn data into structured, visually appealing formats.
- **Communicating Insights and Analysis:** Emphasized presenting data insights through storytelling to make analysis more impactful for stakeholders.

The primary objective of the internship was to develop my ability to use data visualization as a powerful tool to convert complex data into easy-to-understand visuals. This approach aimed to help businesses make more informed decisions. The program not only enhanced my technical skills in creating visuals but also improved my strategic thinking by teaching me how to frame business scenarios and communicate insights effectively.

2. Internship discussion

2.1. Problem Statement Analysis

During my internship project, I was tasked with addressing data visualization challenges for a retail company. The goal was to create insights-driven visuals to help the CEO and CMO make strategic business decisions using tools like Tableau or Power BI. This project required significant data cleanup and the creation of targeted visualizations based on specific questions posed by the company's leadership.

The analysis was divided into several key components:

2.1.1. Data Cleanup

Before diving into the visual analysis, I first conducted a thorough data cleaning process to ensure that the data was accurate and reliable. The following checks were performed:

- **Quantity Check:** I identified and excluded records where the quantity was below 1 unit, indicating returns or data entry errors.
- **Unit Price Check:** I filtered out records with unit prices below \$0, which indicated erroneous data entries.

These data cleanup steps were essential for providing a solid foundation for further analysis and ensuring that the resulting visuals would be based on accurate information.

2.1.2. Creating Visuals for CEO and CMO

Part 1: Time Series Analysis (Question 1)

For the CEO's request, I created a time series visualization that displayed the revenue data for each month in 2011. This visual helped highlight seasonal trends and fluctuations in revenue. This analysis aimed to provide the CEO with insights into monthly performance patterns to help guide future forecasting and planning.

Part 2: Top Countries Analysis (Question 2)

The CMO was interested in identifying the top 10 countries generating the highest revenue, excluding the United Kingdom. I developed a visual representation that combined both revenue and quantity sold for each country. This visual provided the CMO with a clear understanding of the geographic markets that contributed the most to sales, allowing for targeted marketing and sales efforts.

Part 3: Top Customers Analysis (Question 3)

To meet the CMO's request to view the top 10 customers by revenue, I created a descending bar chart visual. This visual placed the highest revenue-generating customers at the top, gradually decreasing to the lower revenue customers. The purpose was to help the CMO identify key customers that contribute significantly to the company's revenue and to ensure that these high-value clients receive focused attention to maintain their satisfaction.

Part 4: Demand Analysis by Region (Question 4)

For the CEO's need to understand product demand by region, I developed a comprehensive map-based visualization. The map displayed demand data for all countries except the United Kingdom, enabling the CEO to pinpoint regions with the highest demand. This analysis was crucial in shaping the company's expansion strategy to tap into new markets with high growth potential.

2.2. Learning Experience

The internship provided me with valuable hands-on experience in data analysis and visualization using Tableau and Power BI. Working on real-world business challenges allowed me to deepen my understanding of data-driven decision-making and develop skills in transforming raw data into actionable insights.

2.2.1. Knowledge Acquired

Throughout the internship, I gained a deeper understanding of data visualization techniques and their impact on strategic business decisions. Some of the key areas of knowledge I acquired include:

- **Data Cleaning Techniques:** Learned how to use conditional logic and data transformation methods to clean and prepare datasets for accurate analysis.
- **Visualization Best Practices:** Enhanced my skills in creating intuitive and insightful visuals that cater to specific business questions using Tableau and Power BI.
- **Storytelling with Data:** Developed the ability to present data in a way that communicates insights effectively to stakeholders, facilitating better decision-making processes.

2.2.2. Skills Learned

The project enabled me to acquire and refine several important skills, including:

- **Analytical Thinking:** Improved my ability to analyze data critically and derive meaningful patterns and trends that align with business objectives.
- **Tool Proficiency:** Gained hands-on experience with Tableau and Power BI, enhancing my ability to create visually compelling dashboards and reports.

- **Problem Solving:** Developed a structured approach to solving data challenges by identifying data issues, cleaning the data, and visualizing insights effectively.

This internship experience has equipped me with both the technical skills and the strategic mindset required to contribute effectively to data-driven decision-making processes in a business environment.

2.2.3. Attitudes and Values Gained

During the job simulation with Tata, I cultivated several essential attitudes and values vital for success in a consulting role:

- **Professionalism:** I recognized the importance of maintaining a professional demeanor while collaborating with team members and presenting insights to stakeholders. This involved clear and concise communication and respect for differing viewpoints.
- **Continuous Learning:** Engaging with tools like Tableau and Power BI emphasized the necessity for ongoing education. I adopted a mindset focused on staying updated with the latest technologies and analytical methods, enhancing my data visualization skills.
- **Team Collaboration:** Working alongside peers reinforced the value of teamwork. I learned that leveraging diverse perspectives fosters innovative solutions and improves decision-making processes.
- **Strategic Thinking:** Addressing real-world challenges for a retail client honed my strategic problem-solving abilities. I focused on aligning data-driven insights with business objectives to provide meaningful recommendations.

2.2.6. Most Challenging Task

One of the most significant challenges during the presentation phase was translating technical data into a compelling narrative that resonated with the leadership team. This task involved :

- **Breaking Down Complex Data:** I had to simplify technical jargon, ensuring that insights were clear and understandable for stakeholders without a technical background.
- **Linking Insights to Business Strategy:** It was crucial to align my data analysis with the company's expansion strategy. This involved highlighting regions with the highest demand potential and formulating recommendations based on revenue trends and customer behavior.
- **Anticipating Leadership Questions:** Understanding that the CEO and CMO would focus on expansion, I prepared to address questions related to lucrative business opportunities, substantiating my answers with data-driven evidence.

This internship experience taught me the importance of not only analyzing data but also effectively communicating insights to support strategic decision-making. I feel more equipped to contribute meaningfully in a consulting role, where data-driven insights are pivotal in shaping business strategies.

3. Conclusion

In conclusion, my internship with Tata, specifically in the area of data visualization, was a highly enriching experience that significantly advanced my understanding of how effective insights can empower businesses. Through the practical tasks I undertook, including framing business scenarios, choosing the right visuals, creating impactful visuals, and communicating insights effectively, I honed my ability to transform raw data into actionable strategies.

This internship provided me with invaluable exposure to real-world applications of data analysis, allowing me to understand the importance of visual storytelling in decision-making processes. The skills I developed in data visualization tools like Tableau and Power BI have equipped me with the technical proficiency necessary to analyze complex datasets and present them in a clear, concise manner.

Additionally, I learned the significance of aligning visual insights with business objectives, enabling me to contribute meaningfully to the strategic goals of an organization. The experience also reinforced the importance of effective communication, ensuring that the insights I presented were not only informative but also persuasive.

As I move forward in my career, I am motivated to leverage the knowledge and skills gained from this internship to drive impactful decisions in future roles. This experience has solidified my passion for data-driven analysis and my commitment to contributing positively to any organization I join.

4. Bibliography

i. About the forage :

<https://www.theforage.com/>

ii. Framing the Business Scenario

<https://www.datapine.com/blog/ceo-dashboard-report-examples-and-templates/>

<https://asana.com/resources/success-metrics-examples>

<https://www.scoro.com/blog/12-business-metrics/>

iii. Choosing the Right Visuals

<https://infogram.com/page/choose-the-right-chart-data-visualization>

<https://towardsdatascience.com/ultimate-guide-to-choosing-the-right-visual-2a77aa8eec08>

<https://chartio.com/learn/charts/how-to-choose-data-visualization/>

<https://www.tableau.com/learn/whitepapers/which-chart-or-graph-is-right-for-you>

iv. Creating Effective Visuals

<https://cdn.theforage.com/vinternships/companyassets/ifobHAoMjQs9s6bKS/5XsFFJu2oCLdmYJW2/1657136958931/Power%20BI%20-%20Tableau%20Downloads.pdf>

<https://cdn.theforage.com/vinternships/companyassets/ifobHAoMjQs9s6bKS/5XsFFJu2oCLdmYJW2/1654309626143/Online%20Retail%20Data%20Set.xlsx>

<https://cdn.theforage.com/vinternships/companyassets/ifobHAoMjQs9s6bKS/5XsFFJu2oCLdmYJW2/1654132626672/Resources%20-%20Tableau%20and%20Power%20BI.pdf>

v. Communicating Insights and Analysis

<https://cdn.theforage.com/vinternships/companyassets/ifobHAoMjQs9s6bKS/5XsFFJu2oCLdmYJW2/1657244683310/Data%20Visualization%20Tableau%20Dashboard%20-%20Empowering%20Business%20for%20effective%20insights.pdf>

<https://insidebigdata.com/2018/03/28/data-scientists-guide-communicating-results/>

<http://elearning.daremightythings.com/CCF/analyzingdata/Print.aspx>

5.Plagiarism Report

